

Policy update on a fail-only trajectory: Vanilla GRPO vs SeRPO

Same failed rollout (final outcome = fail). Each column walks reward → PPO effect on π .

shared trajectory: s_1 explore · s_2 fetch · s_3 premature action (fails) · s_4 retry

✗ Vanilla GRPO (outcome reward)

1 · REWARD — ONE SCALAR FOR THE WHOLE TRAJECTORY

s_1 $r = 0$

s_2 $r = 0$

s_3 $r = 0$

s_4 $r = 0$

↓ PPO update (all rewards equal → no signal)

2 · EFFECT ON POLICY π

s_1 —

s_2 —

s_3 —

s_4 —

All segment rewards are equal (every rollout failed) → **no learning signal**, so PPO leaves π unchanged. The failing step s_3 is never down-weighted; the model repeats the same mistake next time.

π unchanged — nothing learned

✓ SeRPO (segment-level reward)

1 · REWARD — ONE PER SEGMENT

s_1 **5**

s_2 **4**

s_3 **1**

s_4 **4**

↓ PPO update (rewards differ → signal)

2 · EFFECT ON POLICY π

s_1 **$\pi \uparrow$**

s_2 **$\pi \uparrow$**

s_3 **$\pi \downarrow$**

s_4 **$\pi \uparrow$**

Segment rewards differ → PPO **raises** π for the segments that worked and **lowers** π for the failing step s_3 — a targeted update even though the trajectory failed.

π updated — failure step s_3 down-weighted